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Proposal Review 3 : 2601631

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Agency Name:	National Science Foundation
Agency Tracking Number:	2601631
Organization:	Division Of Mathematical Sciences
NSF Program:	ALGEBRA,NUMBER THEORY,AND COM
PI/PD:	Bruce, Juliette
Application Title:	Multigraded Homological Algebra and Geometry
Rating:	Very Good

Review

Summary

Proposal Synopsis

Proposal outlines three main research directions in algebra and geometry with a common threads of multigraded resolutions and regularity.

Intellectual Merit

Strengths

Rating: V

This is one of the most ambitious proposals among those I reviewed. The first project aims to revisit existing ideas (Np conditions, Green's conjecture, Clifford's index, Brill-Noether loci, ar more) in the more general geometric context of stacks. The second project, which revolves around multigraded regularity and free resolutions, would tackle problems of significant interest recent years. The third project strikes me as a novel and interesting approach to algebraic questions arising from toric varieties. PI breaks down the projects into lists of concrete problems work on. PI's previous accomplishments support ability to successfully conduct proposed research.

Weaknesses

Because this is such an ambitious proposal, I think it would benefit the PI to distinguish long term goals from research projects that will be prioritized for the duration of the grant, maybe narrowing the scope a bit and providing a proposed timeline for the work. I think PI could also play up educational value of proposed software for stacks, since it might provide a new way t about a notoriously difficult topic.

Broader Impacts

Strengths

Rating: E

PI has organized several successful conferences, advises undergraduate/graduate students and local student chapters, and has been working with a number of advocacy groups. PI also

develops free open-source software that serves the mathematical community.

Weaknesses

-

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

Summary Statement

A far reaching program motivated by questions of current interest in algebra and geometry. Ranks in the middle third of proposals I reviewed.

Prior NSF support. Intellectual merit includes several research articles that broaden our understanding of homological algebra both in the regular and multigraded setting. Broader impacts include organizing several conferences, various community outreach initiatives, and mentoring undergraduate students.

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